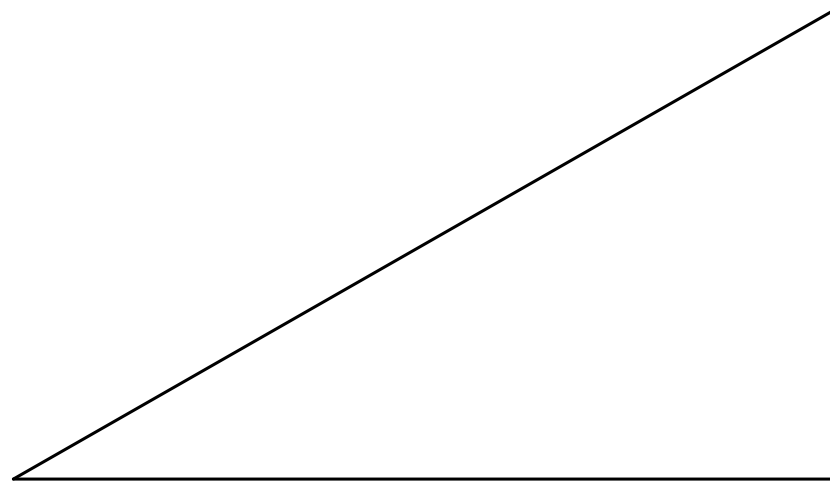




ezdxf

Dibujo automatizado con ezdxf

Ejemplos de la serie



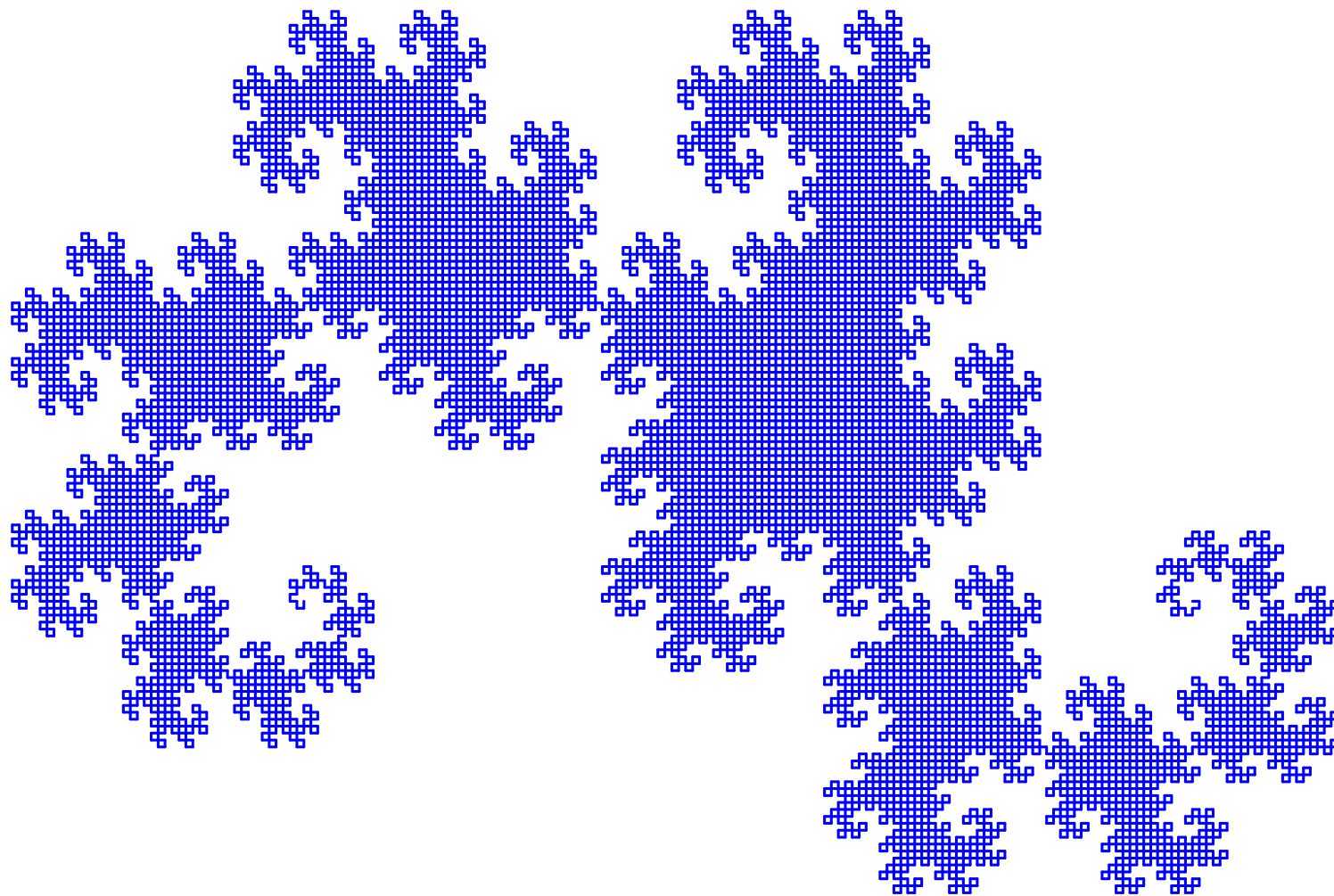
Hola

Hola Mundo

Serie: Dibujo Automatizado con ezdxf

Video 1: Instalación y primeros pasos

EdwinSaul.com

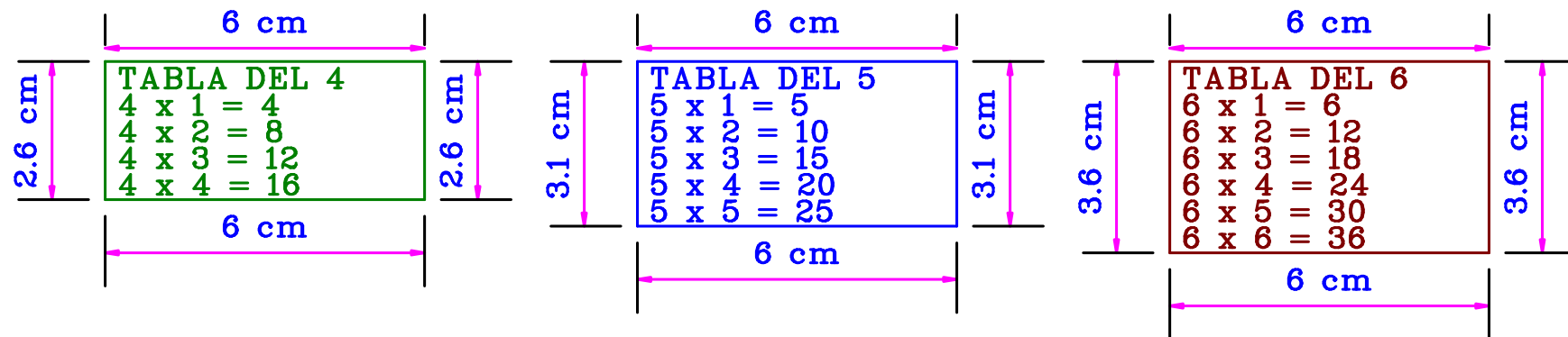


Dragón Fractal

Serie: Dibujo Automatizado con ezdxf

Video 2: Geometrías Básicas

EdwinSaul.com

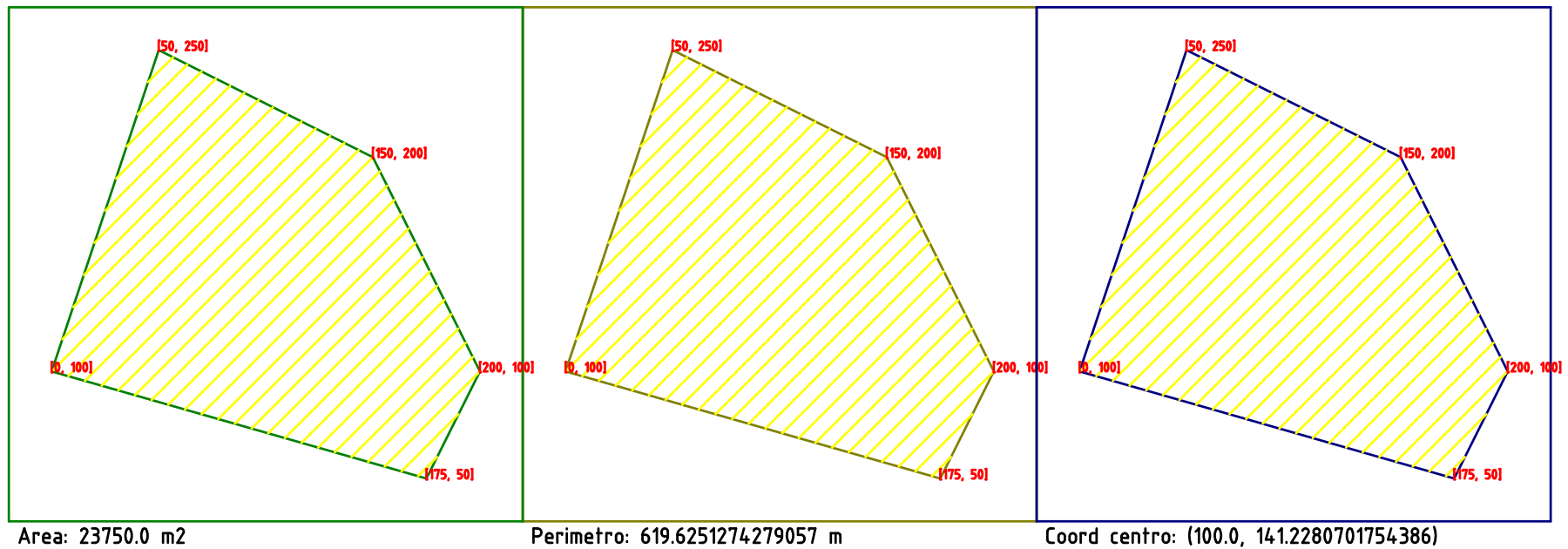


Tablas de Multiplicar

Serie: Dibujo Automatizado con ezdxf

Video 3: Anotaciones y dimensiones

EdwinSaul.com

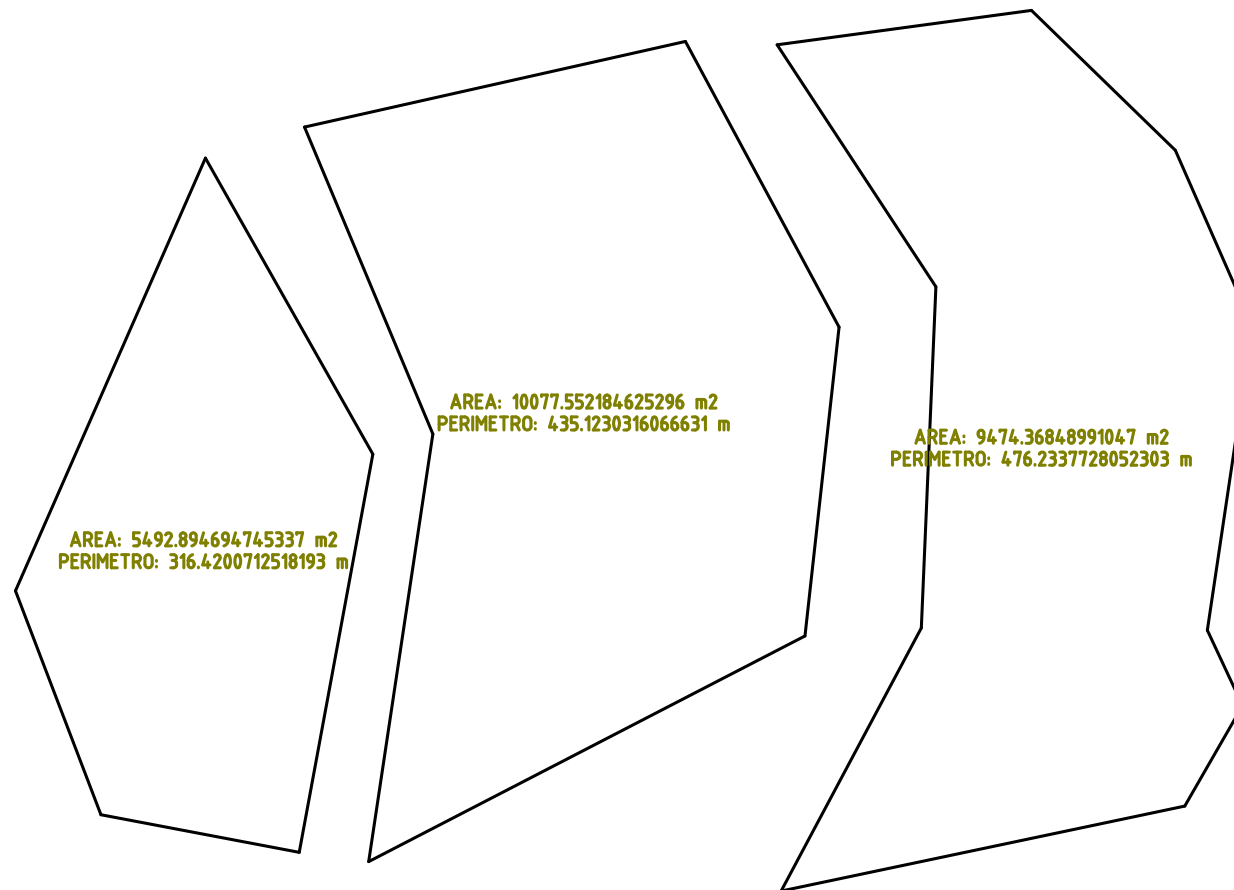


Croquis Perimétrico

Serie: Dibujo Automatizado con ezdxf

Video 4: Elementos del dibujo

EdwinSaul.com

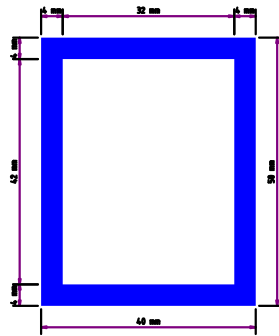


Herramienta de área y perímetro

Serie: Dibujo Automatizado con ezdxf

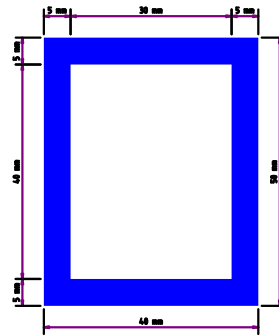
Video 5: Leyendo y analizando archivos DXF

EdwinSaul.com



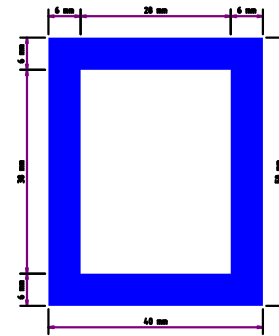
Area = 656
P. ext = 180
P. int = 148

Ancho = 40
Alto = 50
espesor = 4



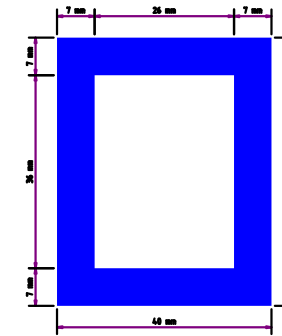
Area = 800
P. ext = 180
P. int = 140

Ancho = 40
Alto = 50
espesor = 5



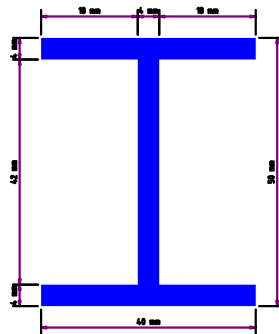
Area = 936
P. ext = 180
P. int = 132

Ancho = 40
Alto = 50
espesor = 6



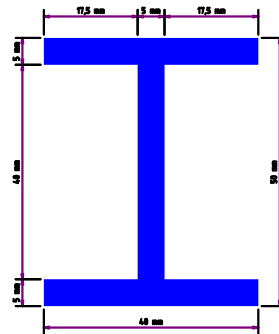
Area = 1064
P. ext = 180
P. int = 124

Ancho = 40
Alto = 50
espesor = 7



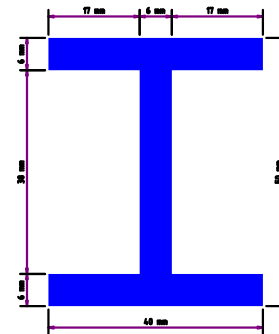
Area = 488
Perim = 252

Ancho = 40
Ala = 4
Alto = 50
Alma = 4



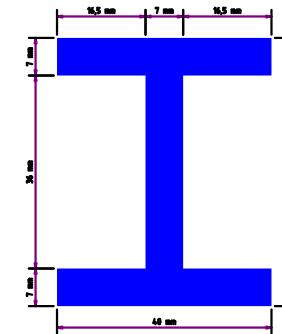
Area = 600
Perim = 250

Ancho = 40
Ala = 5
Alto = 50
Alma = 5



Area = 708
Perim = 248

Ancho = 40
Ala = 6
Alto = 50
Alma = 6



Area = 812
Perim = 246

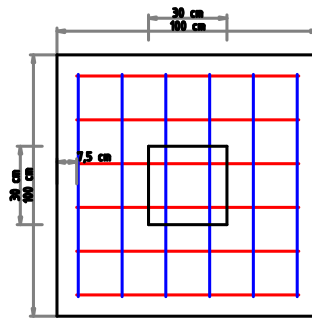
Ancho = 40
Ala = 7
Alto = 50
Alma = 7

Secc.

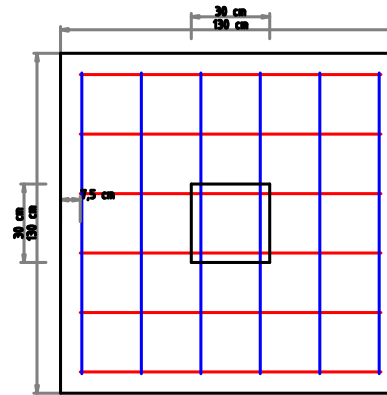
Serie: Dibujo Automatizado con ezdxf

Video 6: programación orientada a objetos y dibujos DXF.

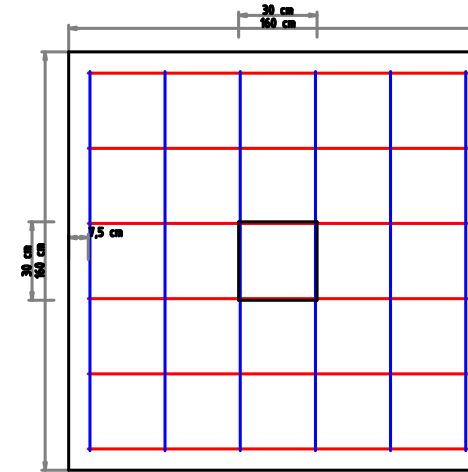
EdwinSaul.com



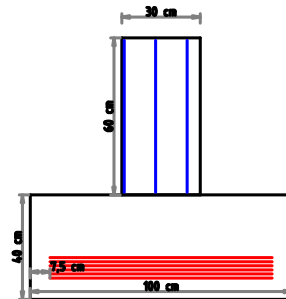
Planta = 100x100 cm
 Columna = 30x30 cm
 Recub = 7.5 cm
 Dir X: 6 barras $\phi=127$ cm
 sep = 17.0 cm $A_s=7.6$ cm²
 Dir Y: 6 barras $\phi=127$ cm
 sep = 17.0 cm $A_s=7.6$ cm²



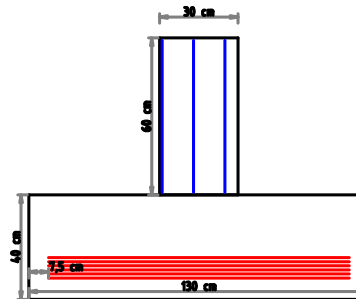
Planta = 130x130 cm
 Columna = 30x30 cm
 Recub = 7.5 cm
 Dir X: 6 barras $\phi=127$ cm
 sep = 23.0 cm $A_s=7.6$ cm²
 Dir Y: 6 barras $\phi=127$ cm
 sep = 23.0 cm $A_s=7.6$ cm²



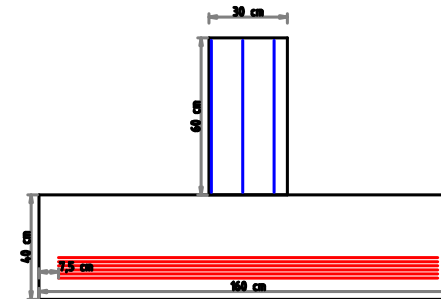
Planta = 160x160 cm
 Columna = 30x30 cm
 Recub = 7.5 cm
 Dir X: 6 barras $\phi=127$ cm
 sep = 29.0 cm $A_s=7.6$ cm²
 Dir Y: 6 barras $\phi=127$ cm
 sep = 29.0 cm $A_s=7.6$ cm²



Zapata = 100x100x40 cm
 Columna = 30x30 cm
 Recub = 7.5 cm
 Barras X: 6 $\phi=127$ cm $A_s=7.6$ cm²
 Barras Y: 6 $\phi=127$ cm $A_s=7.6$ cm²
 A_s total = 15.2 cm²
 Entubo col $\phi=4.5$ cm



Zapata = 130x130x40 cm
 Columna = 30x30 cm
 Recub = 7.5 cm
 Barras X: 6 $\phi=127$ cm $A_s=7.6$ cm²
 Barras Y: 6 $\phi=127$ cm $A_s=7.6$ cm²
 A_s total = 15.2 cm²
 Entubo col $\phi=4.5$ cm

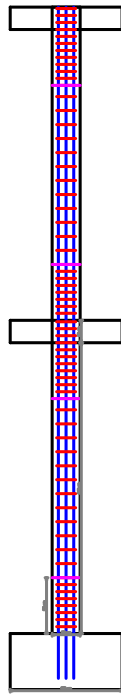


Zapata = 160x160x40 cm
 Columna = 30x30 cm
 Recub = 7.5 cm
 Barras X: 6 $\phi=127$ cm $A_s=7.6$ cm²
 Barras Y: 6 $\phi=127$ cm $A_s=7.6$ cm²
 A_s total = 15.2 cm²
 Entubo col $\phi=4.5$ cm

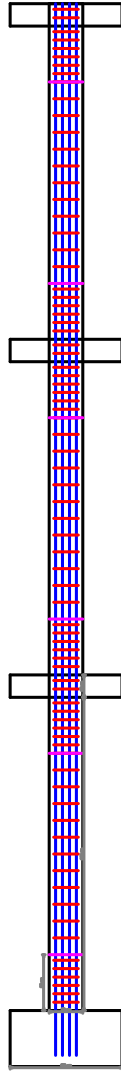
Serie: Dibujo Automatizado con ezdx

Video 7: EJEMPLO 1 - zapatas.

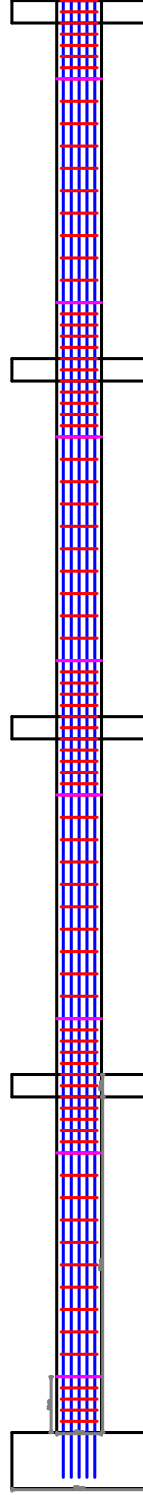
EdwinSaul.com



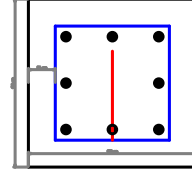
Columna 25x25 cm
 b = 25 cm h = 25 cm
 Ag = 625 cm²
 Recub = 4 cm
 Estribo d = 0.95 cm
 Barras: 8 d=1.27 cm
 (3 en X, 3 en Y)
 As = 10.13 cm²
 Cuantia = 1.62 %
 Sep conf = 6.2 cm
 Sep cent = 12.5 cm



Columna 30x30 cm
 b = 30 cm h = 30 cm
 Ag = 900 cm²
 Recub = 4 cm
 Estribo d = 0.95 cm
 Barras: 12 d=1.59 cm
 (4 en X, 4 en Y)
 As = 23.83 cm²
 Cuantia = 2.65 %
 Sep conf = 7.5 cm
 Sep cent = 15.0 cm



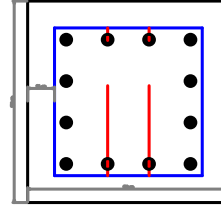
Columna 40x40 cm
 b = 40 cm h = 40 cm
 Ag = 1600 cm²
 Recub = 4 cm
 Estribo d = 0.95 cm
 Barras: 16 d=1.91 cm
 (5 en X, 5 en Y)
 As = 45.84 cm²
 Cuantia = 2.87 %
 Sep conf = 10.0 cm
 Sep cent = 20.0 cm



Seccion columna
 b = 25 cm h = 25 cm
 Ag = 625 cm²
 Recub = 4 cm
 Estribo d = 0.95 cm

Barras: 8 d=1.27 cm
 (3 en X, 3 en Y)
 As = 10.13 cm²
 Cuantia = 1.62 %

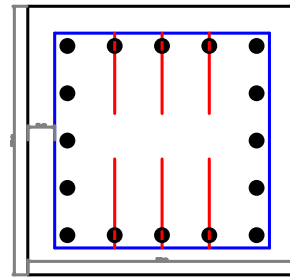
Sep conf = 6.2 cm
 Sep cent = 12.5 cm



Seccion columna
 b = 30 cm h = 30 cm
 Ag = 900 cm²
 Recub = 4 cm
 Estribo d = 0.95 cm

Barras: 12 d=1.59 cm
 (4 en X, 4 en Y)
 As = 23.83 cm²
 Cuantia = 2.65 %

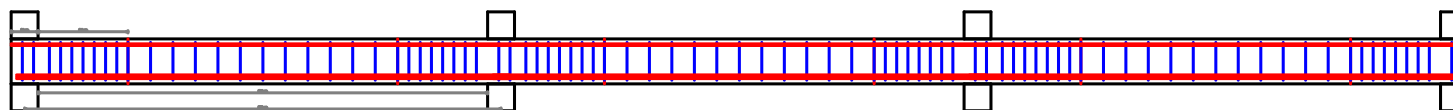
Sep conf = 7.5 cm
 Sep cent = 15.0 cm



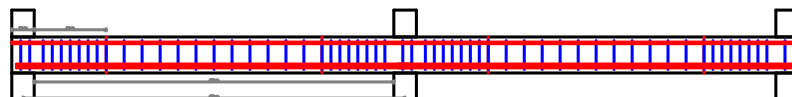
Seccion columna
 b = 40 cm h = 40 cm
 Ag = 1600 cm²
 Recub = 4 cm
 Estribo d = 0.95 cm

Barras: 16 d=1.91 cm
 (5 en X, 5 en Y)
 As = 45.84 cm²
 Cuantia = 2.87 %

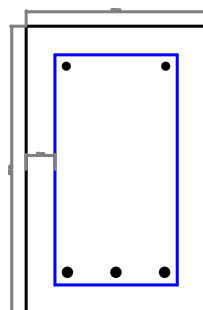
Sep conf = 10.0 cm
 Sep cent = 20.0 cm



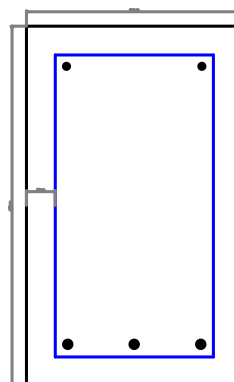
Viga continua 30x50 cm
 f' = 30 MPa
 f_y = 400 MPa
 Recub = 4 cm
 Estribo d = 0.95 cm
 Inf: 3d1.59 As=5.96 cm²
 Sup: 2d1.27 As=2.53 cm²
 Lv = 2h = 100 cm
 s conf = 12.5 cm
 s cent = 1.6 cm



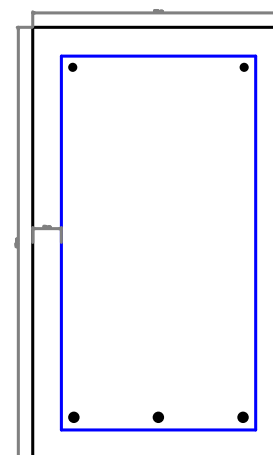
Viga continua 25x40 cm
 f' = 30 MPa
 f_y = 400 MPa
 Recub = 4 cm
 Estribo d = 0.95 cm
 Inf: 3d1.59 As=5.96 cm²
 Sup: 2d1.27 As=2.53 cm²
 Lv = 2h = 80 cm
 s conf = 10.0 cm
 s cent = 1.6 cm



Seccion viga 25x40 cm
 Recub = 4 cm
 Estribo d = 0.95 cm
 Inf: 3d1.59 As=5.96 cm²
 Sup: 2d1.27 As=2.53 cm²
 rho inf = 0.596 %
 Lv = 2h = 80 cm
 s conf = 10.0 cm
 s cent = 1.6 cm



Seccion viga 30x50 cm
 Recub = 4 cm
 Estribo d = 0.95 cm
 Inf: 3d1.59 As=5.96 cm²
 Sup: 2d1.27 As=2.53 cm²
 rho inf = 0.397 %
 Lv = 2h = 100 cm
 s conf = 12.5 cm
 s cent = 1.6 cm

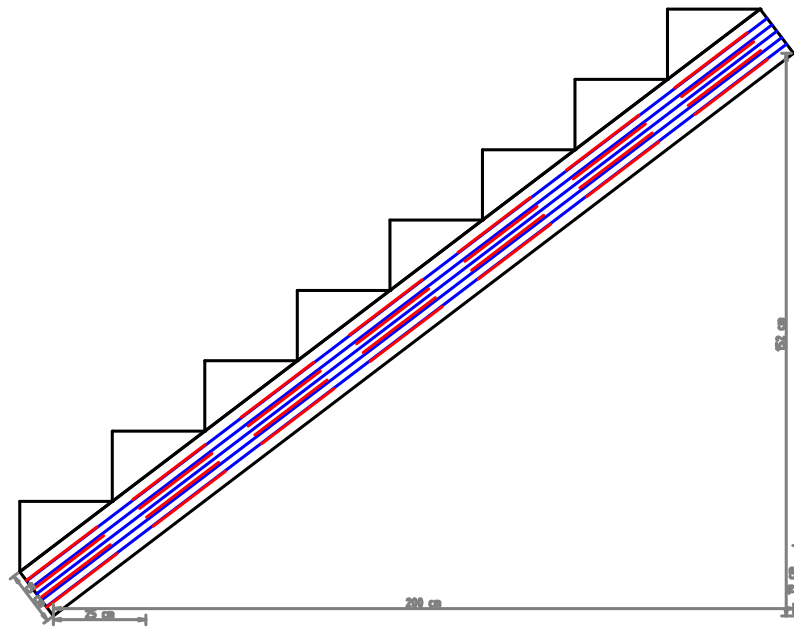


Seccion viga 35x60 cm
 Recub = 4 cm
 Estribo d = 0.95 cm
 Inf: 3d1.59 As=5.96 cm²
 Sup: 2d1.27 As=2.53 cm²
 rho inf = 0.284 %
 Lv = 2h = 120 cm
 s conf = 12.7 cm
 s cent = 1.6 cm

Serie: Dibujo Automatizado con ezdxf

Video 9: EJEMPLO 3 - vigas.

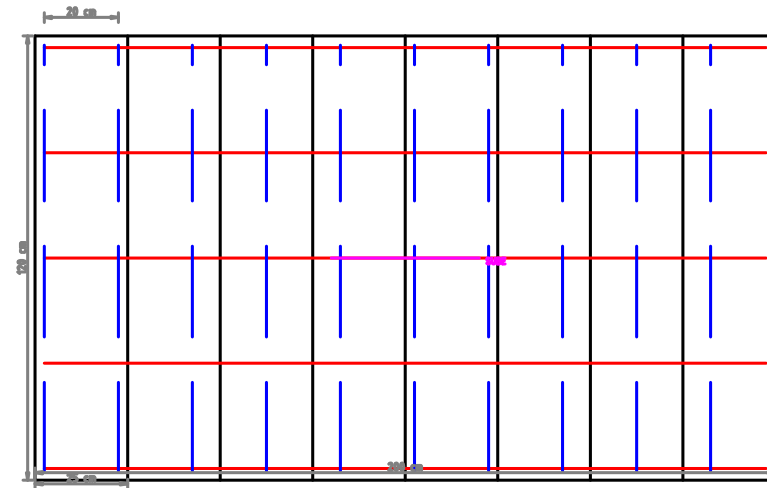
EdwinSaul.com



Escalera de concreto
 N escalones = 8
 Huella = 25 cm
 Contrahuella = 19 cm
 Blondel: $2ch+h = 63$ cm

L horiz = 200 cm
 H total = 152 cm
 Angulo = 37.2 grados
 Ancho = 120 cm
 e losa = 15 cm
 Apoyo = 20 cm

Long: 5d1.27 As=6.334 cm²
 Dist: 4d0.95 As=2.835 cm²



Planta escalera
 N escalones = 8
 Huella = 25 cm
 Contrahuella = 19 cm
 Blondel: $2ch+h = 63$ cm

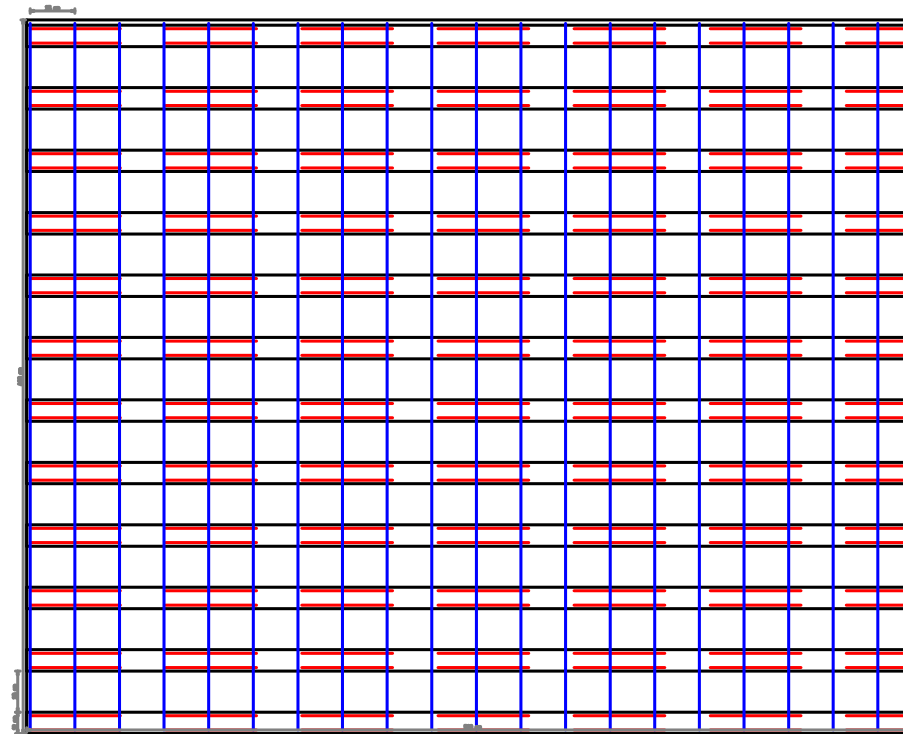
L horiz = 200 cm
 H total = 152 cm
 Ancho = 120 cm
 e losa = 15 cm
 Angulo = 37.2 grados

Long: 5 barras d=1.27 cm
 As=6.334 cm²
 Dist: sep=20 cm N-10
 d=0.95 cm As~7.088 cm²

Serie: Dibujo Automatizado con ezdxf

Video 10: EJEMPLO 4 - escaleras

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Losa aligerada = 500x400 cm
 h total = 17 cm
 Losa esp = 8 cm
 Vigueras = 8 cm
 Recub = 2 cm
 Vigueras h=12 cm
 Recub h=12 cm
 Espo (h-vig) = 35 cm
 h vigueras = 17 cm
 h aligeras = 11 cm
 As top 248.95 = 14.9 cm²
 As bot 248.95 = 14.9 cm²
 As 248.95 = 14.9 cm²
 As 248.95 = 14.9 cm²

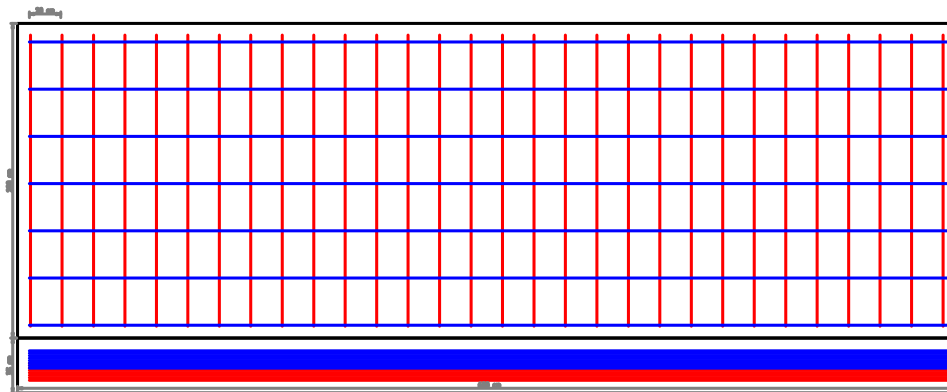


Losa aligerada ancho=400 cm
 h total = 17 cm
 Losa esp = 8 cm
 Vigueras = 8 cm
 Recub = 2 cm
 Vigueras h=12 cm
 Recub h=12 cm
 Espo (h-vig) = 35 cm
 h vigueras = 17 cm
 h aligeras = 11 cm
 As top 248.95 = 14.9 cm²
 As bot 248.95 = 14.9 cm²
 As 248.95 = 14.9 cm²
 As 248.95 = 14.9 cm²

Serie: Dibujo Automatizado con ezdxf

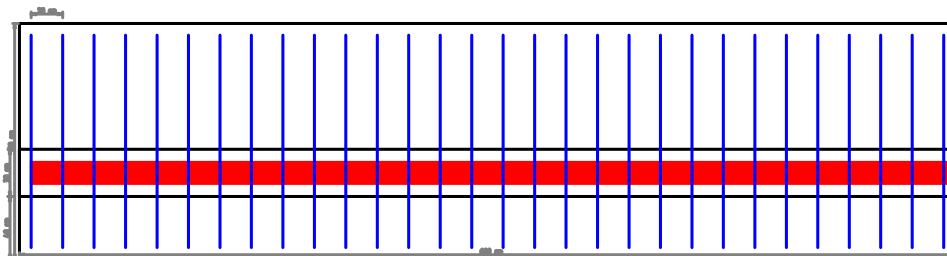
Video 11: EJEMPLO 5 - losas

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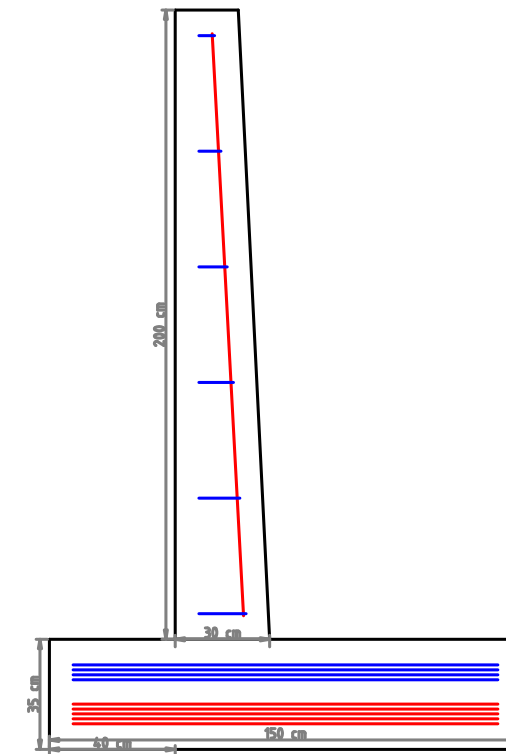
Corte muro L=600 cm
Zapata: 150x35 cm
Pantalla: 600x200 cm e=30 cm
Recub = 7.5 cm

Vert: sep=20 cm N=30 d=1.59
As=59.57 cm²/m
Horiz: sep=30 cm N=7 d=1.27
As=8.87 cm²
Zapata: 8 barras d=1.27
As=10.13 cm²



Planta muro = 600 cm
Zapata: 150 cm hz=35 cm
Puntera=40 cm talon=80 cm
Pantalla: h=200 cm e=30 cm
Recub = 7.5 cm

Long: 10 barras d=1.59 cm As=19.86 cm²
Trans: sep=20 cm d=1.27 cm
N=30 und As=38.0 cm²



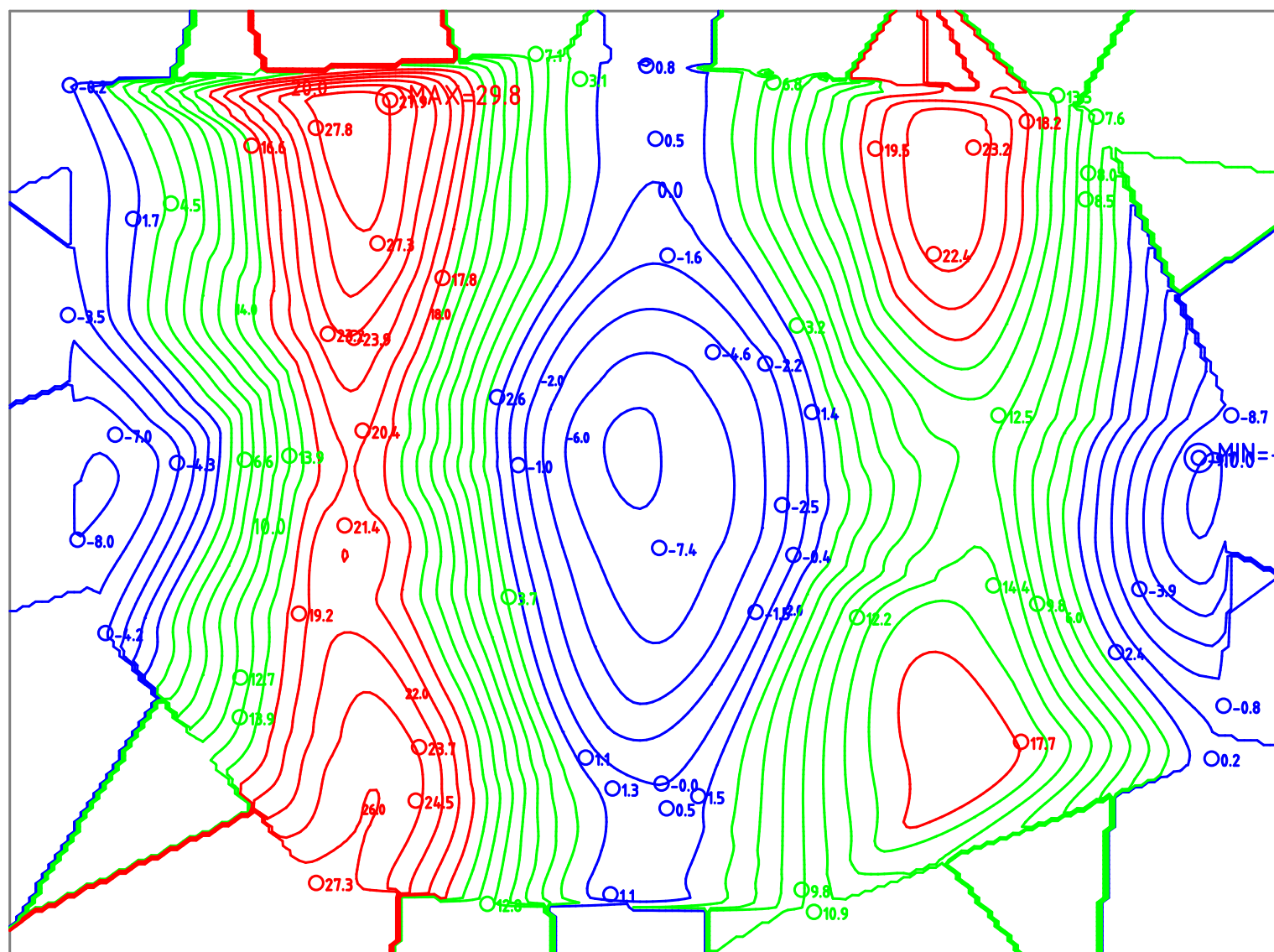
Zapata: 150 cm hz=35 cm
Puntera=40 cm talon=80 cm
Pantalla: h=200 cm
base=30 cm corona=20 cm
Recub = 7.5 cm

Vert: 8 d=1.59 cm As=15.86 cm²
Horiz: 4 d=1.27 cm As=7.6 cm²
Zapata: 5+4 d=1.27 cm As=11.4 cm²

Serie: Dibujo Automatizado con ezdxf

Video 12: EJEMPLO 6 - muro de contención.

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— Zona alta (Z max)
— Zona media
— Zona baja (Z min)

CURVAS DE NIVEL

Z max = 29.8
Z min = -10.6
Rango = 40.4
Equidist = 2.0
N curvas = 20
Maestras c/5
Metodo = cubic
N puntos = 70
Resol = 200x200

Serie: Dibujo Automatizado con ezdx
Video 13: EJEMPLO 7 - curvas de nivel.

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